

Clément Cren

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Position

2023-Present: Hochschulassistent, Beamter (postdoc) at Georg-August Universität Göttingen Mathematisches Institut.

2020-2023: PhD Student at Université Paris-Est Créteil, Laboratoire d'Analyse et de Mathématiques Appliquées. Thesis *Pseudodifferential calculus on filtered manifolds and applications to geometry and harmonic analysis*, supervised by Jean-Marie Lescure.

Education

2019-2020: Master 2 in pure maths at Université de Paris-Cité (former Paris 7); Riemannian geometry, complex geometry, algebraic topology, operator theory, symmetric spaces and Lie groups theory, geometric quantization and geometric analysis (index theory).

Master thesis *Deformation groupoids and inhomogeneous pseudodifferential calculus* supervised by Georges Skandalis at Institut Mathématiques de Jussieu – Paris Rive Gauche (Paris)

July 2019: Successful candidate at Agrégation de Mathématiques, ranked 1st.

2018-2019: Master 2 FESUP (teaching degree) at ENS Cachan, preparation to Agrégation de Mathématiques competitive exam.

2017-2018: Master 1 at ENS Cachan; functional analysis, spectral theory, algebraic number theory, dynamical systems, optimisation, algebraic topology, differential geometry.

Master thesis *Klein metrics on real projective space* supervised by Charles Boubel at Institut de Recherche Mathématique Avancée (Strasbourg)

2016-2017: Bachelor at ENS Cachan; algebra, numerical analysis of ODE&PDE, differential calculus, Fourier analysis, probabilities.

Bachelor thesis *Matrix and tensor completion with applications* supervised by Nicolas Vayatis at Centre de Mathématiques et Leurs Applications (Cachan)

July 2016: Successful candidate at the Ecole Normale Supérieure of Cachan/Paris-Saclay entrance exam.

2014-2016: Classes Prépa in Lycée Chateaubriand Rennes, preparation for french « Grandes Ecoles » competitive entrance exams.

Publications

- Clément Cren, *A transverse index theorem in the calculus of filtered manifolds*, Journal of Functional Analysis, Volume 287, Issue 11, 2024, 110618, ISSN 0022-1236, <https://doi.org/10.1016/j.jfa.2024.110618>.
- C, Cren, *Filtered Calculus and crossed products by R-actions*, 2023, <https://arxiv.org/abs/2308.13532>, submitted to appear in Journal of Operator Theory
- C, Cren, *Transverse parabolic structures and transverse BGG sequences*, <https://www.heldermann.de/JLT/JLT35/JLT351/jlt35009.htm>
- Clément Cren, *Toeplitz algebras and the Heisenberg group*, 2024, <https://arxiv.org/abs/2412.18266>
- Clément Cren, *Stratification of the Helffer-Nourrigat cone*, 2025, <https://arxiv.org/abs/2512.11434>
- Clément Cren, *Fields of Toeplitz algebras for the principal symbol of regular 2-step nilpotent groups*, 2025, <https://arxiv.org/abs/2512.16475>
- Clément Cren & Erfan Rezaei, *Families of Toeplitz operators, family index and deformation quantization*, 2026, <https://arxiv.org/abs/2601.06619>

Teaching Activities

2025-2026: Göttingen Universität. Lecturer for *Algebraic Topology III: Characteristic classes* (42h, Master).

2024-2025: Göttingen Universität. Assistant for *Topological phases and K-theory for Banach algebras* (Bachelor/Master). Lecturer for *LaTeX Kurs*. Assistant for *Algebraic Topology II* (Bachelor/Master). Assistant for *Discrete mathematics for Data Science* (Bachelor).

2023-2024: Göttingen Universität. Assistant for *Funktionalanalysis* (Bachelor/Master). Lecturer for *LaTeX Kurs*. Assistant for *Introduction to algebraic geometry* (Master). Organization of *Onboarding tutorial in Non-commutative Geometry* (Master).

2022-2023: Créteil University. Lecturer for *Introduction to Fourier analysis and wavelets* (Master 2, engineering students); assistant for *Euclidean spaces* (Bachelor, maths students, 2nd year).

2021-2022: Créteil University. Lecturer for *Arithmetic and basic set theory* (Bachelor, maths students, 1st year).

2020-2021: Créteil University. Assistant for *Multivariable calculus and linear algebra* (Bachelor, engineering students, 2nd year). Assistant for *Calculus II* (Bachelor, engineering students, 1st year).

2017-2019: CPGE Lycée Lakanal. Weekly oral examinations.

2016-2017: Tutoring for bachelor in science and technology students at Institut Villebon-Charpak (University Paris-Saclay)

Supervision

2026: Göttingen Universität. Supervision of a Bachelor thesis of a student from Université Paul Sabatier (Toulouse). Subject (temporary) : *Representations of Clifford algebras and spin groups*.

2025: Göttingen Universität. Supervision of a Bachelor thesis of a student from Université Paul Sabatier (Toulouse). Subject: *Kirillov orbit method and stratification of the space of co-adjoint orbits*. Supervision of a Master thesis of a student from ENS Rennes (co-supervision with Thomas Schick). Subject: *Obstruction to Fredholm perturbations on manifolds with boundary*.

2021: Créteil University. Supervision of two Master 1 Thesis *Differential Forms and Poincaré Lemma*

Awards

2024: Laureate of [PhD thesis award from COMUE Paris-Est-Sup](#)

July 2019: Successful candidate at Agrégation de Mathématiques, ranked 1st.

Communications

Talks in conferences, workshops, seminars

March 2026: Toulouse, Géométrie et Topologie seminar, <https://indico.math.cnrs.fr/category/473/>

December 2025: Albi, Workshop 2025 ANR OpArt,
<https://indico.math.cnrs.fr/event/15013/overview>

July 2025: Würzburg, Workshop Singular foliations and higher structures,
<https://www.mathematik.uni-wuerzburg.de/en/geometry/research/events/single/news/workshop-singular-foliations-and-higher-structures-in-wuerzburg/>

June 2025: Lund, Hypoellipticity in Lund conference.
<https://www.maths.lu.se/english/research/seminars/hypoellipticity-in-lund-2025/>

June 2025: Potsdam, Analysis seminar

April 2025: Toulouse, Regards croisés sur l'hypoellipticité conference, organised by RT Tétraèdre,
<https://indico.math.cnrs.fr/event/13702/overview>

March 2025: Angers, *Probabilités, Physique Mathématique et Analyse (2PMA)* seminar,
<https://math.univ-angers.fr/seminaires/2pma/>

February 2025: Clermont-Ferrand, *Géométrie, Algèbre, Algèbres d'opérateurs* seminar,
https://lmbp.uca.fr/seminaires/gt_gaao.php

November 2024: Besse, ANR OpART workshop, <https://indico.math.cnrs.fr/event/12234/>

November 2024: Brest, *Géométrie et Topologie* seminar, <https://www.univ-brest.fr/laboratoire-mathematiques-bretagne-atlantique/fr/page/seminaire-geometrie-et-topologie>

June 2024: Cortona, *Noncommutative Geometry and applications* conference, <https://sites.google.com/view/ngc2024cortona>

May 2024: Paris, ANR OpART launching meeting, <https://indico.math.cnrs.fr/event/11553/>

April 2024: CIRM, *Group operator algebras and Non commutative geometry* conference, <https://conferences.cirm-math.fr/2987.html>

March 2024: Sichuan University Chengdu, *Algebraic, analytic, geometric structures emerging from quantum field theory* school and workshop, <https://math.scu.edu.cn/info/1031/10971.htm>

December 2023: Université de Lorraine Metz, GDR Géométrie Non-Commutative *Arbre de Noël* conference, <https://iecl.univ-lorraine.fr/gnc2023/>

November 2023: Forum des jeunes mathématicien-nes *Analyse. Géométrie. Applications.*, <https://forumbxl2023.sciencesconf.org/#:~:text=Le%20Forum%20des%20jeunes%20math%C3%A9maticien,22%20au%2024%20novembre%202023.>

June 2023: Université de Lorraine Metz, *Théorie de Lie, Géométrie et Analyse* seminar, <https://iecl.univ-lorraine.fr/evenements/seminaires/seminaire-theorie-de-lie-geometrie-et-analyse/>

March 2023: Athens, *Noncommutative Geometry and Representation Theory* conference, <https://conferences.uoa.gr/event/56/>

September 2022: Université Paris-Cité, Operator Algebras seminar, <https://www.imj-prg.fr/gestion/evenement/affSeance/8831>

June 2022: Cortona, *Dirac Operators in Topology, Geometry, and Representation Theory* conference, <https://cortona2022.cbaer.eu/>

June 2022: Caen, Géométrie Non-Commutative GDR Conference, <https://conf.lmno.cnrs.fr/GNC2022/>

May 2022: Dartmouth College, Functional Analysis Seminar

Other attended conferences and schools

Décembre 2025: Albi, RT Tétraèdre *Arbre de Noël de Géométrie Non-Commutative* conference, <https://indico.math.cnrs.fr/event/15053/>

Novembre 2025: Reims, Workshop on Mathematical Physics and Pseudo-differential operators – Celebrating Jean Nourrigat’s 80th birthday, <https://indico.math.cnrs.fr/event/13798/>

September 2023: Angers, *High frequency analysis: from operator algebras to PDEs* workshop, <https://indico.math.cnrs.fr/event/9728/>

July 2023: Köln, *Quantization in geometry* conference, http://www.mi.uni-koeln.de/geometrische_analysis/QCONF_2023/Ohne-Titel.html

March 2023: Göttingen, *Interactions between Poisson geometry and quantization* spring school, <https://www.uni-goettingen.de/en/672633.html>

December 2022: IHES, GDR Géométrie Non-Commutative *Arbre de Noël* conference, <https://indico.math.cnrs.fr/event/8849/>

March 2022: CIRM, *Geometry and Analysis on non-compact manifolds* conference, <https://conferences.cirm-math.fr/2548.html>

June 2021: Oberwolfach Workshop *Analysis, Geometry and Topology of singular PDE*, <https://publications.mfo.de/handle/mfo/3883>

Organization

October 2026: Organisation of an Autumn school, *C*-Algebras, representations and dynamics* School (tentative title) at Göttingen Universität, supported by RTG 2491

September 2026: Organisation of a conference *Pseudodifferential techniques in singular analysis* (tentative title) at Göttingen Universität.

October 2025: Organisation of an Autumn school *Spectra of Analysis in Geometry and Physics*, at Göttingen Universität, supported by RTG 2491, <https://www.uni-goettingen.de/en/698852.html>

Science Outreach

2022-2023: Scientific advisor for the MATH.en.JEANS association in collège Janson de Sailly, <https://www.mathenjeans.fr/content/College-Janson-de-Sailly-Paris-2022-2023>

Languages

French: Native

English: Fluent (C1 Cambridge Advanced certified)

Spanish: Intermediate level

German: Beginner

Programming Languages

Latex: Fluent

Python: Intermediate level

Ocaml: basic level